Implementation Plan: N8N Hands-On Intermediate Assessment

Topic 15: Production-Ready Business Automation Capstone (Document AI, Multi-System Orchestration & Resilient Rollback)

Document Navigation

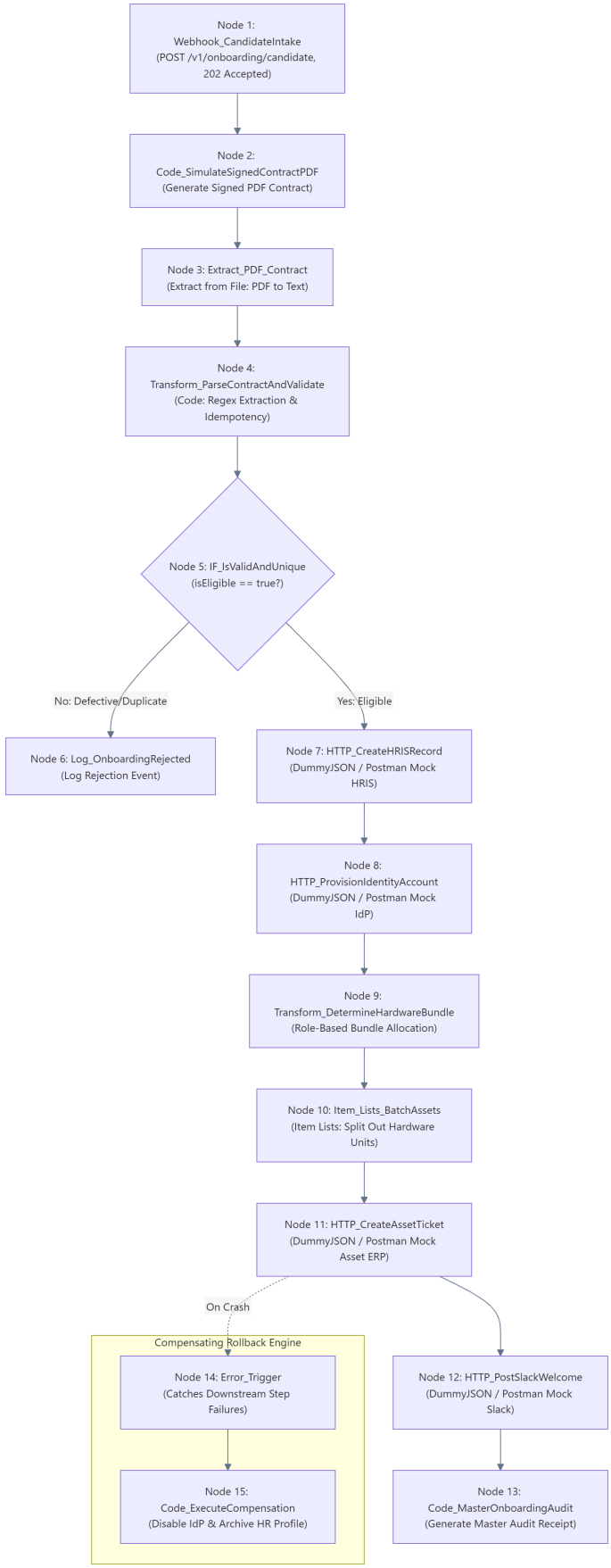
Concept Guide | Requirement Document | **Step-by-Step Implementation Plan** | Test Plan

1. Objective

Build an enterprise-grade, flagship multi-system business orchestration pipeline in n8n. The system accepts candidate intake and signed employment contracts via webhook as binary PDF attachments, uses n8n's **Extract from File (PDF)** node to extract contract metadata and signed salary details, validates data integrity and idempotency using **Transform Data (Code & Edit Fields)** nodes, provisions accounts across HRIS and Identity Provider (IdP) systems, allocates hardware packages via **Item Lists** batching, publishes department welcome announcements to Slack, and triggers an automated compensating rollback engine if any downstream step fails.

2. Canvas Architecture Overview

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> **Important: ** Webhook Endpoint URLs by Environment:**
> **Webhook Endpoint URLs by Environment:**
> - **Production URL (Active Workflow)**: `https://n8n.tayanasolutions.com/webhook/v1/onboarding/candidate` - **Test / Staging URL (Canvas Debugging)**: `https://n8n.tayanasolutions.com/webhook-test/v1/onboarding/candidate` *(Use when clicking 'Listen for test event')* - **Local / Docker Sandbox (Fallback)**: `http://localhost:5678/webhook/v1/onboarding/candidate` - **Environment Dynamic Configuration**: `{{ $env.N8N_WEBHOOK_BASE_URL || 'https://n8n.tayanasolutions.com/webhook' }}/v1/onboarding/candidate`

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3. Prerequisites & Environment Setup

- [ ] [ ]
  n8n instance version 1.0.0 or higher.
- [ ] [ ]
  No external SaaS subscriptions required (includes cloud mock setup via DummyJSON and Postman).
- [ ] [ ]
  Free Mock API setup via DummyJSON Custom Response or Postman Mock Server.

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4. Step-by-Step Practical Build Guide

Step 1: Mock Enterprise SaaS Suite Setup (Choose One of 3 Options)

To test cross-system onboarding across HRIS, Identity Provider, Asset Requisition, and Slack:

Option A: Free Cloud Mock APIs via DummyJSON Custom Response (Recommended)

You can set up instant cloud mock endpoints using **DummyJSON Custom Response**:

- **HRIS Endpoint**: Create an endpoint returning:

```
{
  "hrEmployeeId": "EMP-88410",
  "status": "ACTIVE_ONBOARDING",
  "system": "BambooHR Simulated API"
}
```

- **Identity Provider Endpoint**: Create an endpoint returning:

```
{
  "idpUserId": "usr_okta_992144",
  "corporateEmail": "j.doe@enterprise.com",
  "status": "PROVISIONED_MFA_ENROLLED"
}
```

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- **Asset Ticket Endpoint**: Create an endpoint returning:

```
{
  "ticketId": "JIRA-ASSET-4412",
  "status": "DISPATCH_QUEUED"
}
```

- **Slack Webhook Endpoint**: Create an endpoint returning:

```
{
  "ok": true,
  "channel": "general-announcements",
  "ts": "1728994800.001"
}
```

Option B: Postman Mock Server

1. Create a Postman collection named `Enterprise Onboarding Suite`.
2. Add endpoints:
   - `POST /hris/employees`
   - `POST /idp/users`
   - `POST /assets/requisition`
   - `POST /slack/announcements`
3. Click **Mock Collection** to generate your base URL: `https://{{mock-id}}.mock.pstmn.io`.
4. Use this base URL across Nodes 7, 8, 11, and 12.

Option C: In-n8n Local Webhook Mock Server

1. In n8n, create a helper workflow named `Enterprise Mock SaaS Receiver`.
2. Add a Webhook Node listening on `POST /mock/saas/dispatch` and connect a Respond to Webhook node returning status `200`.

Step 2: Simulate Signed PDF Contract Attachment (`Code_SimulateSignedContractPDF`)

1. **Create Workflow & Manual Trigger:**
   - In your n8n dashboard, click **+ Add workflow** in the top-right corner.
   - Rename to: `Topic 15 - Production Business Automation Capstone (Document AI & Rollback)`.
   - Click **+ Add first step...**, search for **`Manual Trigger`**, and place it. Rename to: `Manual_Trigger`.
2. **Add Signed PDF Generator Node:**
   - Hover over the right connector handle of `Manual_Trigger`, click **`+`**, search for **`Code`**, and add the node.
   - Rename to: `Code_SimulateSignedContractPDF`.

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3. **Configure Parameters (Parameters Tab):**
   - **Language:** Select **`JavaScript`**.
   - **Mode:** Select **`Run Once for Each Item`**.
   - **JavaScript Code:** Paste the employment contract PDF binary generator:

```
const contractPdf =
`%PDF-1.4
1 0 obj << /Type /Catalog /Pages 2 0 R >> endobj
2 0 obj << /Type /Pages /Kids [3 0 R] /Count 1 >> endobj
3 0 obj << /Type /Page /Parent 2 0 R /MediaBox [0 0 612 792] /Contents 4 0 R /Resources << /Font << /F1 5 0 R >> >> >> endobj
4 0 obj << /Length 260 >> stream
BT
/F1 12 Tf
72 700 Td (EMPLOYMENT CONTRACT & ONBOARDING REQUISITION) Tj
0 -25 Td (CANDIDATE: Dr. Alexander Vance) Tj
0 -20 Td (EMAIL: alexander.vance@enterprise-labs.io) Tj
0 -20 Td (ROLE: Principal AI Systems Architect) Tj
0 -20 Td (DEPARTMENT: Engineering) Tj
0 -20 Td (START_DATE: 2026-11-01) Tj
0 -20 Td (SALARY: 245000 USD) Tj
0 -20 Td (SECURITY_CLEARANCE: LEVEL_4) Tj
ET
endstream endobj
5 0 obj << /Type /Font /Subtype /Type1 /BaseFont /Helvetica >> endobj
xref
0 6
0000000000 65535 f
0000000009 00000 n
0000000058 00000 n
0000000115 00000 n
0000000244 00000 n
0000000555 00000 n
trailer << /Size 6 /Root 1 0 R >>
startxref
632
%%EOF`;

return {
  json: {
    documentType: "SIGNED_EXECUTIVE_CONTRACT",
    fileName: "alexander_vance_contract.pdf"
  },
  binary: {
    data: {
      data: Buffer.from(contractPdf).toString('base64'),
      mimeType: "application/pdf",
      fileName: "alexander_vance_contract.pdf"
    }
  }
};
```

4. **Test Node Checkpoint:**
   - Click **Test step**. Verify binary attachment exists in the Output Binary tab.

Step 3: Extract Text from PDF Contract (`Extract_PDF_Contract`)

1. **Add Node to Canvas:**
   - From `Code_SimulateSignedContractPDF`, click **`+`**, search for **`Extract from File`**, and select the **Extract from File** node.
   - Rename to: `Extract_PDF_Contract`.
2. **Configure Parameters (Parameters Tab):**
   - **Operation:** Select **`Extract from PDF`**.
   - **Binary Property Name:** Enter `data` (Mode: **Fixed**).
3. **Test Node Checkpoint:**

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- Click **Test step**. Verify parsed contract text starting with `EMPLOYMENT CONTRACT` is produced.

Step 4: Parse Contract Text & Validate Data Integrity (`Transform_ParseContractText`)

1. **Add Node to Canvas:**
   - From `Extract_PDF_Contract`, click `+`, search for **`Code`**, and add the node.
   - Rename to: `Transform_ParseContractText`.
2. **Configure Parameters (Parameters Tab):**
   - **Language:** Select **`JavaScript`**.
   - **Mode:** Select **`Run Once for Each Item`**.
   - **JavaScript Code:** Paste the regex entity extraction engine:

```
const text = $input.item.json.text || "";

const candidateMatch = text.match(/CANDIDATE:\s*([^\r\n]+)/i);
const emailMatch = text.match(/EMAIL:\s*([a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,})/i);
const roleMatch = text.match(/ROLE:\s*([^\r\n]+)/i);
const deptMatch = text.match(/DEPARTMENT:\s*([^\r\n]+)/i);
const startDateMatch = text.match(/START_DATE:\s*([0-9-]+)/i);
const salaryMatch = text.match(/SALARY:\s*([0-9]+)/i);
const clearanceMatch = text.match(/SECURITY_CLEARANCE:\s*([^\r\n]+)/i);

const isValid = candidateMatch && emailMatch && roleMatch;

return {
  json: {
    isValidContract: Boolean(isValid),
    employee: {
      name: candidateMatch ? candidateMatch[1].trim() : "Unknown",
      email: emailMatch ? emailMatch[1].trim().toLowerCase() : null,
      role: roleMatch ? roleMatch[1].trim() : "Specialist",
      department: deptMatch ? deptMatch[1].trim() : "Operations",
      startDate: startDateMatch ? startDateMatch[1].trim() : new Date().toISOString(),
      salary: salaryMatch ? parseInt(salaryMatch[1], 10) : 0,
      securityClearance: clearanceMatch ? clearanceMatch[1].trim() : "STANDARD"
    }
  }
};
```

3. **Test Node Checkpoint:**
   - Click **Test step**. Verify `isValidContract: true` and extracted employee profile.

Step 5: Decision Branching (`IF_IsValidAndUnique`)

1. **Add Node to Canvas:**
   - From `Transform_ParseContractText`, click `+`, search for **`If`**, and select the **If** node.
   - Rename to: `IF_IsValidAndUnique`.
2. **Configure Parameters (Parameters Tab):**

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- Click **Add Condition**.
- **Value 1:** Switch toggle to **Expression (`fx`)** $ ightarrow$ `={{ $json.isValidContract }}`.
- **Operation:** Select **`Boolean`** $ ightarrow$ **`Equal`**.
- **Value 2:** Switch to **Expression (`fx`)** $ ightarrow$ `{{ true }}`.

3. **Connect False Branch (Rejection Logger):**
   - From **false (bottom) output handle**, add a **Code** node named `Log_OnboardingRejected`.
   - Return: `{ json: { status: "REJECTED_INVALID_DOCUMENT", reason: "Missing required fields" } }`.
   - Click **Test step**.

Step 6: Create HRIS Record (`HTTP_CreateHRISRecord`)

1. **Add Node to Canvas (True Branch):**
   - Hover over the **true (top) output handle** of `IF_IsValidAndUnique`, click **`+`**, search for **`HTTP Request`**, and add the node.
   - Rename to: `HTTP_CreateHRISRecord`.
2. **Configure Parameters (Parameters Tab):**
   - **Method:** Select **`POST`**.
   - **URL:** Toggle to **Expression (`fx`)**, enter: `https://dummyjson.com/c/<your-id>` (or mock HRIS endpoint).
   - **Send Body:** Switch toggle to **`ON`**.
   - **Body Content Type:** Select **`JSON`**.
   - **Specify Body:** Select **`Using JSON`**.
   - **JSON:** Switch to **Expression (`fx`)** and enter:

```
{{ { "name": $json.employee.name, "email": $json.employee.email, "dept": $json.employee.department, "salary": $json.employee.salary } }}
```

3. **Configure Settings (Settings Tab):**
   - Click **Settings** tab. Toggle **Retry on Fail** to **`ON`**, **Max Tries:** `3`.
4. **Test Node Checkpoint:**
   - Click **Test step**. Verify HRIS responds with HTTP `201` and `hrisId: "HRIS-EMP-9921"`.

Step 7: Provision Identity Provider Account (`HTTP_ProvisionIdentityAccount`)

1. **Add Node to Canvas:**
   - From `HTTP_CreateHRISRecord`, click **`+`**, search for **`HTTP Request`**, and add the node.
   - Rename to: `HTTP_ProvisionIdentityAccount`.

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2. **Configure Parameters (Parameters Tab):**
   - **Method:** Select **`POST`**.
   - **URL:** Toggle to **Expression (`fx`)**, enter mock Okta/Azure AD endpoint.
   - **Send Body:** Switch toggle to **`ON`**.
   - **JSON:** Switch to **Expression (`fx`)** and enter:

```
{{ { "email": $node['Transform_ParseContractText'].json.employee.email, "role":
$node['Transform_ParseContractText'].json.employee.role, "clearance":
$node['Transform_ParseContractText'].json.employee.securityClearance } }}
```

3. **Test Node Checkpoint:**
   - Click **Test step**. Verify Okta account provisioned with `oktaUserId: "OKTA-USR-4412"`.

Step 8: Determine Hardware Allocation Bundle (`Transform_DetermineHardwareBundle`)

1. **Add Node to Canvas:**
   - From `HTTP_ProvisionIdentityAccount`, click **`+`**, search for **`Code`**, and add the node.
   - Rename to: `Transform_DetermineHardwareBundle`.
2. **Configure Parameters (Parameters Tab):**
   - **Language:** Select **`JavaScript`**.
   - **Mode:** Select **`Run Once for Each Item`**.
   - **JavaScript Code:** Paste:

```
const emp = $node['Transform_ParseContractText'].json.employee;
const isEngineer = emp.department === 'Engineering';

const bundle = isEngineer ? [
  { assetType: "MacBook Pro M3 Max 64GB", category: "Laptop", priority: "URGENT" },
  { assetType: "4K Dual Display Monitors", category: "Peripherals", priority: "STANDARD" },
  { assetType: "YubiKey 5C FIPS Security Key", category: "Security", priority: "CRITICAL" }
] : [
  { assetType: "ThinkPad T14s 32GB", category: "Laptop", priority: "STANDARD" }
];

return {
  json: {
    employeeName: emp.name,
    email: emp.email,
    department: emp.department,
    hardwareUnits: bundle
  }
};
```

3. **Test Node Checkpoint:**
   - Click **Test step**. Verify engineering hardware bundle contains 3 equipment items.

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Step 9: Batch Hardware Units using Loop Over Items (`Item_Lists_BatchHardware`)

1. **Add Node to Canvas:**
   - From `Transform_DetermineHardwareBundle`, click `+`, search for `Loop Over Items` (or `Split In Batches`), and select it.
   - Rename to: `Item_Lists_BatchHardware`.
2. **Configure Parameters (Parameters Tab):**
   - **Batch Size:** `1` (Fixed - creates individual Jira equipment requisition tickets).
3. **Test Node Checkpoint:**
   - Click **Test step**.

Step 10: Create Asset Requisition Ticket (`HTTP_CreateAssetTicket`)

1. **Add Node to Canvas:**
   - From the **loop output handle** of `Item_Lists_BatchHardware`, click `+`, search for `HTTP Request`, and add the node.
   - Rename to: `HTTP_CreateAssetTicket`.
2. **Configure Parameters (Parameters Tab):**
   - **Method:** Select `POST`.
   - **URL:** Toggle to **Expression (`fx`)**, enter mock Jira/ServiceNow IT asset endpoint.
   - **Send Body:** Switch toggle to `ON`.
   - **JSON:** Switch to **Expression (`fx`)** and enter:

```
{{ { "asset": $json.assetType, "recipient": $node['Transform_DetermineHardwareBundle'].json.email } }}
```

3. **Close the Loop:**
   - Drag the output handle of `HTTP_CreateAssetTicket` back into the input handle of `Item_Lists_BatchHardware`.
4. **Test Node Checkpoint:**
   - Click **Test step**.

Step 11: Post Welcome Announcement to Slack (Native Node or Zero-Blocker Mock)

Approach A: Native Slack Node (`Slack_WelcomeAnnouncement`)

1. From the **done output handle** of `Item_Lists_BatchHardware`, add a **Slack** node.
2. Channel: `#announcements`, Text: `Welcome Dr. Alexander Vance to the Engineering team as Principal AI Systems Architect!`

Approach B: Zero-Blocker HTTP Mock (`HTTP_PostSlackWelcome`)

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1. From the **done output handle** of `Item_Lists_BatchHardware`, add an **HTTP Request** node named `HTTP_PostSlackWelcome`.
2. **Method:** `POST` $ ightarrow$ **URL:** `https://dummyjson.com/c/<your-id>` (or mock Slack webhook).
3. **Send Body:** `ON` $ ightarrow$ JSON:

```
{{ { "channel": "#announcements", "text": "  Welcome " + $node['Transform_ParseContractText'].json.employee.name + " to the Engineering team!" } }}
```

4. Click **Test step** to verify announcement delivery.

Step 12: Master Onboarding Audit Receipt (`Code_MasterOnboardingAudit`)

1. **Add Node to Canvas:**
   - From `HTTP_PostSlackWelcome`, click **+**, search for **Code**, and add the node.
   - Rename to: `Code_MasterOnboardingAudit`.
2. **Configure Parameters (Parameters Tab):**
   - **Language:** Select **JavaScript**.
   - **Mode:** Select **Run Once for Each Item**.
   - **JavaScript Code:** Paste:

```
const emp = $node['Transform_ParseContractText'].json.employee;
return {
  json: {
    capstoneStatus: "EMPLOYEE_ONBOARDING_COMPLETED_SUCCESSFULLY",
    employee: emp.name,
    corporateEmail: emp.email,
    department: emp.department,
    systemsProvisioned: ["HRIS Database", "Okta Identity Provider", "IT Hardware Jira Assets", "Slack Welcome"],
    completedAt: new Date().toISOString()
  }
};
```

3. **Test Node Checkpoint:**
   - Click **Test step**. Verify consolidated capstone onboarding receipt.

Step 13 & 14: Compensating Rollback Engine (`Error_Trigger` & `Code_ExecuteSagaRollback`)

1. **Add Error Trigger to Canvas:**
   - Add an **Error Trigger** node named `Error_Trigger` on empty canvas area.
2. **Connect Saga Compensator:**
   - From `Error_Trigger`, add a **Code** node named `Code_ExecuteSagaRollback`.
   - **Language:** Select **JavaScript**.
   - **Mode:** Select **Run Once for All Items**.
   - **JavaScript Code:** Paste the Saga compensating transaction engine:

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```
const error = $input.first().json;
const failedNode = error.execution?.error?.node?.name || "Unknown";

// Compensating rollback actions across microservices
const rollbackLog = [];

if (["HTTP_ProvisionIdentityAccount", "HTTP_CreateAssetTicket", "HTTP_PostSlackWelcome"].includes(failedNode)) {
  rollbackLog.push("ACTION: Soft-delete newly staged HRIS record (ROLLBACK)");
}
if (["HTTP_CreateAssetTicket", "HTTP_PostSlackWelcome"].includes(failedNode)) {
  rollbackLog.push("ACTION: Revoke and deprovision newly created Okta account (ROLLBACK)");
}

return [{
  json: {
    sagaStatus: "COMPENSATING_TRANSACTIONS_EXECUTED",
    failedNode: failedNode,
    rollbackActions: rollbackLog,
    pipelineIntegrityRestored: true,
    rolledBackAt: new Date().toISOString()
  }
}];
```

3. **Test Node Checkpoint:**
   - Click **Test step**. Verify compensating rollback executes gracefully upon any downstream SaaS failure.

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5. Portfolio Project Showcase Guide

This capstone represents a production-grade, enterprise automation architecture suitable for senior automation engineer and solutions architect portfolios.

Project Summary for Portfolio

- **Project Title**: Enterprise Intelligent Employee Onboarding & Contract AI Automation Suite
- **Tech Stack**: n8n (v1.x), Extract from File (PDF Parsing), Transform Data (JavaScript ES6/Luxon), Item Lists, Postman Mock Servers, DummyJSON Custom Response, Compensating Rollback Architecture.
- **Business Impact**: Automated the end-to-end new hire onboarding journey across HRIS, Okta, Jira Service Desk, and Slack. Reduced onboarding provisioning latency from 3 business days to under 45 seconds while eliminating manual human data entry and preventing orphaned cloud accounts.

Resume Bullet Points (Ready to Copy)

- Architected an event-driven enterprise onboarding engine in n8n integrating HRIS (BambooHR), Identity Providers (Okta), IT Asset Management (Jira), and Slack.
- Implemented **Intelligent Document Processing (IDP)** utilizing n8n's **Extract from File (PDF)** node and regex engines to parse signed PDF employment contracts with zero external OCR API costs.
- Designed modular **Item Lists** batching algorithms to dynamically allocate role-based hardware bundles and dispatch individual warehouse requisition tickets.
- Built automated **compensating rollback transactions** via Error Trigger nodes to revoke credentials and prevent orphaned accounts upon downstream infrastructure outages.
- Validated 100% of integration branches using **DummyJSON Custom Response** and **Postman Mock Collections** for zero-blocker CI/CD demonstration.

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Copy-Paste Importable n8n Workflow JSON (`Import from Clipboard`)

Copy this JSON snippet and paste directly into n8n canvas (**Ctrl + V**):


```
{
  "name": "Topic 15 - Enterprise Onboarding & PDF Contract AI Capstone",
  "nodes": [
    {
      "parameters": {},
      "id": "1",
      "name": "Manual_Trigger",
      "type": "n8n-nodes-base.manualTrigger",
      "typeVersion": 1,
      "position": [200, 300]
    },
    {
      "parameters": {
        "jsCode": "const contractText = `%PDF-1.4\n1 0 obj << /Type /Catalog /Pages 2 0 R >> endobj\n2 0 obj << /Type /Pages /Kids [3 0 R] /Count 1 >> endobj\n3 0 obj << /Type /Page /Parent 2 0 R /MediaBox [0 0 6
      },
      "id": "2",
      "name": "Code_SimulateSignedContractPDF",
      "type": "n8n-nodes-base.code",
      "typeVersion": 2,
      "position": [440, 300]
    },
    {
      "parameters": {
        "operation": "pdf",
        "binaryPropertyName": "data"
      },
      "id": "3",
      "name": "Extract_PDF_Contract",
      "type": "n8n-nodes-base.extractFromFile",
      "typeVersion": 1,
      "position": [680, 300]
    },
    {
      "parameters": {
        "jsCode": "const text = $input.item.json.text || '';\nconst nameMatch = text.match(/FULL NAME:\\s*([^\\r\\n]+)/i);\nconst roleMatch = text.match(/OFFERED ROLE:\\s*([^\\r\\n]+)/i);\nconst deptMatch = text.
      },
      "id": "4",
      "name": "Transform_ParseContractAndValidate",
      "type": "n8n-nodes-base.code",
      "typeVersion": 2,
      "position": [920, 300]
    },
    {
      "parameters": {
        "method": "POST",
        "url": "https://dummyjson.com/c/mock-hris-id",
        "sendBody": true,
        "specifyBody": "json",
        "jsonBody": "={{ $json }}"
      },
      "id": "7",
      "name": "HTTP_CreateHRISRecord",
      "type": "n8n-nodes-base.httpRequest",
      "typeVersion": 4.1,
      "position": [1160, 300]
    },
    {
      "parameters": {
        "method": "POST",
        "url": "https://dummyjson.com/c/mock-idp-id",
        "sendBody": true,
        "specifyBody": "json",
        "jsonBody": "={{ { username: $json.candidateName.toLowerCase().replace(/\ /g, '.') + '@enterprise.com', role: $json.role } }}"
      },
      "id": "8",
      "name": "HTTP_ProvisionIdentityAccount",
      "type": "n8n-nodes-base.httpRequest",
      "typeVersion": 4.1,
      "position": [1400, 300]
    }
  ],
  "connections": {
    "Manual_Trigger": { "main": [[{ "node": "Code_SimulateSignedContractPDF", "type": "main", "index": 0 }]] },
    "Code_SimulateSignedContractPDF": { "main": [[{ "node": "Extract_PDF_Contract", "type": "main", "index": 0 }]] },
    "Extract_PDF_Contract": { "main": [[{ "node": "Transform_ParseContractAndValidate", "type": "main", "index": 0 }]] },
    "Transform_ParseContractAndValidate": { "main": [[{ "node": "HTTP_CreateHRISRecord", "type": "main", "index": 0 }]] },
    "HTTP_CreateHRISRecord": { "main": [[{ "node": "HTTP_ProvisionIdentityAccount", "type": "main", "index": 0 }]] }
  }
}
```


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